



Regional differentiation of foliar N/Ca in *Fagus americana* and *F. sylvatica* on acid soils

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ABSTRACT

Prolonged periods of air pollution depleted soil Ca and increased N supply in forests of Eastern North America and Central Europe. We compared a cool rural to a warm urban forest site to explore the geographic range of foliar N, Ca, and N/Ca on naturally regenerated saplings of American beech and European beech in Nova Scotia (Canada) and Saarland (Germany), respectively. Foliar N/Ca of beech saplings remained variable due to geo-climatic gradients and bio-climatic site factors, tight internal cycling of Ca, and varying pollution history. Our data suggest that deep and extreme soil Ca depletion renders climatic factors the drivers of foliar N/Ca in beech forests. Global warming will therefore amplify the potential risks of changing nutrient supply ratios in regions with a prolonged history of severe acid rain. Existing forest site classifications usually capture regional differentiation of climatic factors but often fail to assess impacts of acid rain history on foliar Ca that will be required for regional differentiation of foliar nutrient ratios on acid soils. However, homogenization of foliar Ca can be expected in regions with a history of severe acid rain.

1. Introduction

The two congeneric species American beech (*Fagus americana* Ehrh.; AB) and European beech (*F. sylvatica* L.; EB) are common throughout the deciduous forests of eastern North America and central Europe, respectively. The two regions have been exposed for decades to acidifying N and S emissions from anthropogenic sources severely depleting exchangeable base cation pools of non-calcareous soils most notably Ca. Contrary to the depletion of base cations soils have become enriched and saturated with N in severely polluted regions (Meesenburg et al., 2016; Verstraeten et al., 2017).

Repeated inventories of foliar N and Ca revealed temporal divergences between the two nutrients for EB in northeastern France (Duquesnay et al., 2000) although the ratio of foliar N to base cations did not increase significantly during long-term monitoring of severely polluted forest sites in central Germany and Flanders (Meesenburg et al., 2016; Verstraeten et al., 2017). Foliar N increased for AB along an N pollution gradient in the Southern Appalachian (Boggs et al., 2005). Within a geographically more narrowly defined area of the southern Saarland (Germany), Bouman (2014) found close linear relationships for foliar N vs. chlorophyll across calcareous and non-calcareous soils while a nutritional difference between the two soil types became evident in the foliar N/Ca ratio of EB. Bouman (2014) suggested that foliar

N was primarily linked to phenology-driven chlorophyll formation thereby overriding geochemical effects on foliar N within the study area. The geochemical N/Ca contrast reported by Bouman (2014) was in good agreement with a long-term field experiment in Normandie (France) in which lime application increased foliar Ca of EB but not N (Forey et al., 2015). The same was observed for AB following lime application in Quebec (Canada; Ouimet et al., 2017). Moreover, the dynamics of natural biological processes in plants and soils are well known to be temperature-dependent. Temperature is obviously controlled by the geographic situation of a forest within a region (i.e. geo-climatic factor) and modulated at the site level by shading due to forest canopy closure (i.e., bio-climatic factor). Shading slows down N mineralization in the soil (Bartsch, 2000) and also limits the development of N-rich tissues in leaves (Bresinsky et al., 2008).

The purpose of this paper is to explore the geographic range of foliar N/Ca on saplings of the common temperate tree genus *Fagus* at distinct juxtapositions of air pollution and climate gradients at the regional scale in Nova Scotia (Canada) and Saarland (Germany). In each of the two regions, we compared beech saplings in advanced regeneration (i.e., beneath forest canopy) at a cool rural to a warm urban forest on well drained acid soils. We expected the divergence between N and Ca (i.e., foliar N/Ca) to increase from low N availability in cool rural forests to high N availability in warm urban forests due to the positive

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effect of temperature on N-cycling as well urban areas continue to be hot spots of N pollution. We also expected a more pronounced site effect on N/Ca for AB at the periphery of the continental acid rain belt in Eastern North America than for the congeneric EB at the centre of the acid rain belt in Europe. Thus, we hypothesized regional homogenization of foliar Ca indicative of profound Ca depletion due to severe pollution history.

2. Materials and methods

2.1. Study sites

In the Province of Nova Scotia on the Atlantic coast of Canada, we selected a deciduous rural hardwood near East Bay (46°N; 60°01' W; 100 m a.s.l.) and a deciduous urban green belt close to high vehicle traffic close to the capital city Halifax (44.65°N; 63°45' W; 80 m a.s.l.) for the testing of foliar N, Ca, and N/Ca on advanced natural regeneration of AB. Early county-level surveys by MacDougall et al. (1963) and Cann et al. (1963) described the soils of the two forest sites as stony and shallow with pH 4.0 near soil surface increasing to 5.2 below 0.3 m depth. The authors also reported an increase of the frost-free period from 137 to 171 days that corresponds to a normal climate (1980–2010) with a mean temperature of 5.9 °C and an annual precipitation of 1517 mm for the cool rural and 7.5 °C and 1468 mm for the warm urban forest, respectively (<http://climate.weather.gc.ca>). In the German Saarland, we selected a sub montane beech forest north of rural Nonnweiler (49.39°N; 7°02' E; 600 m a.s.l.) and a beech forest close to the capital city Saarbrücken (49°15'N 7°00'E; 300 m a.s.l.) for testing EB saplings. The two forest sites correspond to a mean temperature increase from 8.0 to 9.6 °C and a decrease of precipitation from 1050 to 700 mm in the Saarland (Anonymous, 1987). Anonymous (1987) also reported stable top soil pH values of 3.7 in 1973 and 3.8 in 1982 for the shallow stony soils derived from Taunus quartzite in the cool rural forest; and, for the warm urban forest a pH of 4.4 in the early 1960s and pH 4.3 in 1982 for sandy soils derived from Triassic sandstone. However, Anonymous (1987) reported dramatic decreases of soil Ca contents for the two eco-districts in Saarland at stable soil pH. Environmental maps show higher rates of acidifying nitrate deposition for the warm urban than for the cool rural forest in Nova Scotia (Anonymous, 2003) and much higher rates for the Saarland compared to Nova Scotia (Charlier, 2012). Recent air quality monitoring also indicated higher ambient NO₂ concentrations for Saarbrücken compared to Nonnweiler as well as Halifax (Anonymous, 2015; <https://novascotia.ca/nse/airdata>).

2.2. Data collection

In late summer, we randomly selected a total of 12 saplings of naturally regenerated beech in each of the four forests, to manually clip from each sapling a total of 18 leaves along proximal to distal and bottom to top crown positions. The AB leaves were collected in 2015 while EB was sampled in the warm urban forest in 2013 and in the cool rural forest in 2016. Blades were separated from petioles, dried at 70 °C for three days, digested in nitric acid for eight hours, filtered, and then analyzed for Ca using atomic absorption spectrometry. Foliar N was determined using a LECO analyzer. We also determined the specific leaf mass (i.e., mass of dried leaves per unit leaf area) of sapling foliage as an indicator of light exposure (or shading) thus bio-climatic characterization of the forest sites (Petriřan et al., 2009). Four leaves were collected from four different crown positions (i.e., top proximal, top distal, mid proximal, and mid distal). Leaf area was determined gravimetrically on photocopied leaves (Bouman et al., 2012). For soil sampling, we used a small shovel to gradually remove the organic surface layer of litter and partially decomposed plant material (i.e., O_L and O_F) and to excavate soil at four equidistant spots around each sapling to a depth of 0.05 m. The four sub samples were mixed for

determination of soil in the lab. Soil samples were saturated with distilled water, stirred, and allowed to equilibrate prior to measuring the pH using a pH meter calibrated in standard buffer solutions at pH 7 and 4. In Nova Scotia, we also submitted our soil samples to the Nova Scotia Department of Agriculture Laboratory for analysis of total soil N and loss on ignition (LOI). The LOI was also determined on the cool rural sample collected in Saarland.

2.3. Data analysis

Data were organized in MS Excel to compute for each sample means, standard errors of means, data ranges (i.e., minimum and maximum values) and to conduct statistical tests. Homogeneity of variance between samples was ascertained using the F-test prior to performing t-tests. If the F-test failed (even following logarithmic transformations) at $p < 0.05$, we selected the *t*-test option assuming unequal variance. The *p*-value of the *t*-test indicates the probability that two sample means differ due to chance alone.

3. Results and discussion

3.1. American beech in Nova Scotia

Comparable levels of soil acidification for the cool rural and warm urban forest in Nova Scotia are in good agreement with the original county surveys dating back more than five decades (Table 1; MacDougall et al., 1963; Cann et al., 1963). Mean pH values remained within the Al buffer range (i.e., pH 3.8–4.2) typically characterized by low soil Ca availability in relation to annual Ca uptake by the forest thus tight internal cycling of Ca in northern hardwoods of Atlantic and Eastern Canada (Chen et al., 2013; Bouman, 2018). On average, the accumulation of soil organic matter (i.e., LOI) was much higher at the cool rural than warm urban site although the two data sets each showed a high variance and the site effect was statistically non-significant ($p = 0.259$). The SLM of AB saplings differed however, significantly between the two forests ($p = 0.034$; Table 2). The difference indicates increased shading of saplings beneath the mature tree canopy of the cool rural forest. Conversely, an SLM increase at the warm urban site indicates enhanced soil N supply and formation of more N-rich leaf tissue (Bartsch, 2000; Bresinsky et al., 2008). Thus, the significant increase of AB foliar N from cool rural to warm urban ($p = 0.001$) is consistent with a concordance of the geo-climatic and bio-climatic gradient in Nova Scotia. The foliar N value for AB in the cool-rural forest compares well with AB of the northern hardwood region in Quebec (Ouimet et al., 2017). The elevated foliar N range in the warm-

Table 1

pH, total N, and loss on ignition (LOI) of top soil for beech saplings naturally regenerated in cool rural ($n = 12$) and warm urban ($n = 12$) forests of Nova Scotia (Canada) and Saarland (Germany).

Site		Amer. beech (Nova Scotia)			Eur. beech (Saarland)		
		Mean	min.	max.	Mean	min.	max.
pH	cool rural	4.2 (0.1)	3.8	4.7	3.2 (0.1)	2.9	3.4
	warm	4.1 (0.0)	3.9	4.4	3.6 (0.1)	3.4	4.1
	urban	<i>0.547</i>			<i>0.000</i>		
N _{tot} [%]	cool rural	0.61 (0.08)	0.27	1.36	–	–	–
	warm	0.51 (0.10)	0.25	1.46	–	–	–
	urban	<i>0.421</i>					
LOI [%]	cool rural	29.4 (4.2)	13.8	68.5	23.1 (1.9)	14.7	79.0
	warm	22.7 (4.0)	11.7	56.8	–	–	–
	urban						

Note: Values in parentheses denote standard errors of means; italicized values denote *p*-values for *t*-tests.

Table 2

Specific leaf mass (SLM) and foliar nutrients for beech saplings naturally regenerated in cool rural (n = 12) and warm urban (n = 12) forests of Nova Scotia (Canada) and Saarland (Germany).

Site		Amer. beech (Nova Scotia)			Eur. beech (Saarland)		
		Mean	min.	max.	Mean	min.	max.
SLM [gm ⁻²]	cool rural	28.1 (1.2)	21.9	35.0	34.5 (1.8)	27.4	47.2
	warm urban	32.5 (1.5) <i>0.034</i>	25.8	39.9	22.5 (0.4) <i>0.000</i>	20.7	25.6
N [mgg ⁻¹]	cool rural	20.9 (0.5)	18.4	23.3	23.3 (1.4)	15.6	32.1
	warm urban	24.8 (0.9) <i>0.001</i>	18.3	29.4	27.7 (0.5) <i>0.010</i>	25.4	30.1
Ca [mgg ⁻¹]	cool rural	6.8 (0.4)	4.1	8.6	1.8 (0.2)	0.6	2.9
	warm urban	5.0 (0.5) <i>0.006</i>	2.3	6.6	2.0 (0.1) <i>0.620</i>	1.4	2.6
N/Ca [gg ⁻¹]	cool rural	3.2 (0.3)	2.57	5.7	15.7 (2.3)	7.0	28.4
	warm urban	5.5 (0.6) <i>0.002</i>	3.57	11.2	14.6 (0.9) <i>0.632</i>	10.8	20.3

Note: Values in parentheses denote standard errors of means and italicized values denote *p*-values for t-tests.

urban forest corresponds however, to values reported for AB along an N pollution gradient in the Southern Appalachian (Boggs et al., 2005). Significantly low foliar Ca for AB at the warm urban site in Nova Scotia was likely caused by Ca depletion of the soil profile due to exposure to elevated levels of acid rain (Anonymus, 2003). Our value range for foliar Ca falls within the range previously reported for a large sample of AB in Nova Scotia four decades ago (Langille and MacLean, 1976) and across the eastern deciduous forest of North America (Boggs et al., 2005; Ouimet et al., 2017; Richardson, 2017). A significant differentiation of N/Ca in AB saplings at the regional scale (*p* = 0.002) supports the first hypothesis that foliar N/Ca increases from cool-rural to warm-urban forest sites.

3.2. European beech in Saarland

Soil acidification has progressed in the Saarland during the past three decades as indicated by lower pH values in the two forests compared to early soil testing reported by Anonymous (1987) although the site difference has remained significant (Table 1). The persistent site difference likely reflects an inherent acid buffering capacity effect due to a low reactive soil mass in the shallow and stony soils derived from Taunus quartzite in the cool rural forest. However, very low foliar Ca in each of the two Saarland forests (Table 2) corroborates our second hypothesis of homogenized Ca nutrition of EB at the regional scale due to a history of severe acid rain and concomitant Ca depletion of non-calcareous forest soils (Gerber et al., 2001). As opposed to geochemical homogenization of foliar Ca due to air pollution, foliar N significantly increased in EB from cool rural to warm urban (*p* = 0.010). The expected increase of foliar N due to the geo-climatic gradient was however, moderated in the Saarland by a significant decrease of the SLM in the urban EB saplings (*p* < 0.001; Table 2). The SLM decrease indicated more shading and less light exposure which in turn exerted a negative bio-climatic effect on soil N mineralization and formation of N-rich leaf tissue at the warm urban site (Bartsch, 2000; Bresinsky et al., 2008). Apparently, the bio-climatic effect counter acted the geo-climatic effect on foliar N to the extent that the regional differentiation of the N/Ca ratio became statistically non-significant (*p* = 0.632) in the Saarland. However, high foliar N and dramatically low foliar Ca values raise concerns about the release of natural N constraints on food webs and herbivory in terrestrial ecosystems (Meunier et al., 2016). Positive effects of N saturation on herbivory might be amplified by profound Ca depletion given the important role of Ca in xylogenesis and cell wall construction in woody plants (Fromm, 2010).

4. Conclusion

Nutrient ratios such as foliar N/Ca remain variable due to geographic variation of inherent bio-climatic site factors, tight internal cycling of Ca on acid soils, and varying pollution history. Our data suggest that deep and extreme Ca depletion in regions with a history of severe acid rain may render regional climate gradients and site-specific bio-climatic factors the drivers of foliar N/Ca in advanced regeneration of beech forests. Global warming will therefore amplify the potential risks of changing nutrient supply ratios in regions with profound soil Ca depletion due to a prolonged history of severe acid rain. Existing forest site classifications usually capture climatic factors but often fail to assess the impacts of acid rain history on foliar Ca that will be required for regional differentiation of foliar nutrient ratios on acid soils. However, homogenization of foliar Ca can be expected in regions with a history of severe acid rain. Moreover, silviculture can moderate foliar N/Ca by selective tree felling that will favour forest canopy closure, low ground temperature, and low SLM compared to clear cut and shelter wood systems.

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